7.8 First Order Schemes

Let us consider a scheme of the form $[\![\mathrm{XYZ}]\!]$, where each letter in the string is distinct and corresponds to one of the A, B or O Langevin splitting pieces (giving six possible schemes). Given the simple structure of the scheme, we expect to recover a first order error in our invariant (long-time) averages. By composing the semigroup operators

$$\exp(h\mathcal{L}^\dagger_{[\![\mathrm{XYZ}]\!]}) = \exp(h\mathcal{L}^\dagger_{\mathrm{Z}}) \exp(h\mathcal{L}^\dagger_{\mathrm{Y}}) \exp(h\mathcal{L}^\dagger_{\mathrm{X}}),$$

and computing the terms in the BCH expansion, we have

$$\hat{\mathcal{L}}^\dagger_{[\![\mathrm{XYZ}]\!]} = \mathcal{L}^\dagger_{\mathrm{LD}} + h\mathcal{L}^\dagger_{1,[\![\mathrm{XYZ}]\!]} + \mathcal{O}(h^2),$$

where

$$\mathcal{L}^\dagger_{1,[\![\mathrm{XYZ}]\!]} = -\frac{1}{2}\left([\mathcal{L}^\dagger_{\mathrm{X}}, \mathcal{L}^\dagger_{\mathrm{Y}}] + [\mathcal{L}^\dagger_{\mathrm{X}}, \mathcal{L}^\dagger_{\mathrm{Z}}] + [\mathcal{L}^\dagger_{\mathrm{Y}}, \mathcal{L}^\dagger_{\mathrm{Z}}]\right). \tag{7.20}$$

The first order correction function for this scheme is then given by the particular solution to the PDE,

$$\mathcal{L}^\dagger_{\mathrm{LD}}\left(f_{1,[\![\mathrm{XYZ}]\!]}\rho_\beta\right) = -\mathcal{L}^\dagger_{1,[\![\mathrm{XYZ}]\!]}\rho_\beta. \tag{7.21}$$

The general solution of (7.21) will consist of the sum of the homogeneous solution $f^{[\mathrm{HS}]}_{1,[\![\mathrm{XYZ}]\!]}(\boldsymbol{q},\boldsymbol{p})$ and particular solution $f^{[\mathrm{PS}]}_{1,[\![\mathrm{XYZ}]\!]}(\boldsymbol{q},\boldsymbol{p})$ in the usual way. However, since $\mathcal{L}^\dagger_{\mathrm{LD}}\vartheta = 0$ has as its only solution $\vartheta \propto \rho_\beta$, the homogeneous solution of (7.21) is

$$f^{[\mathrm{HS}]}_{1,[\![\mathrm{XYZ}]\!]}(\boldsymbol{q},\boldsymbol{p}) = c,$$

for some constant c. Our assumption that $\langle f_{1,[\![\mathrm{XYZ}]\!]}\rangle = 0$ means that $c = -\langle f^{[\mathrm{PS}]}_{1,[\![\mathrm{XYZ}]\!]}\rangle$, and hence we only require a particular solution $f^{[\mathrm{PS}]}_{1,[\![\mathrm{XYZ}]\!]}(\boldsymbol{q},\boldsymbol{p})$.

Thus for any first order scheme our line of attack will be to compute the perturbation operator $\mathcal{L}^\dagger_1$, plug it into the right hand side and solve the PDE to find the suitable correction function. It turns out that this is not so difficult at leading order, for simple first order schemes such as this. However at higher orders the complexity of the terms often limits our progress.